Task 7 – User Stories & Use Cases

Project: EGYPT-MEDIA-VAULT | **Owner:** Mohamed Khaled (7.1) with Ibrahim Mowafy (7.2) | **Reviewer:** Mustafa Mahmoud | **Status:** Final draft for review, handoff 10/13/2026

Inputs: Task 5 KPI Catalog, Task 6 Stakeholder Analysis, Project Proposal. **Outputs to:** Functional Requirements; UI/UX Design & Prototyping.

1. Scope, Actors and Terminology

Actors (aligned with the Task 6 register). Task 6 lists the Project Team as one group and uses 'system maintainer' in User Story 4. Here the team is split into two roles so stories and use cases match.

Actor	Type	Role
BI User / Data Analyst	Human	Views dashboards and reports, compares sources, analyzes KPIs.
System Administrator	Human	Monitors pipeline runs, investigates failures, manages the source list.
Data Engineer	Human	Owns data quality: validation, standardization, duplicate handling, story matching rules.
Scheduler	Supporting system	Triggers collection and processing runs automatically.
Egyptian News Websites	External systems	Provide articles and publication metadata.

General News Readers are indirect stakeholders (no direct interface in this version). Their need is met through the insights BI Users produce, so they have no stories of their own.

Glossary. *Source* = one news website. *Ingested article* = an article collected and stored in the warehouse (KPI01 term; Task 6 uses 'collected'). *Story group* = a set of articles from different sources judged to cover the same or a similar story (Task 5 calls this a story cluster). *Run* = one execution of collection and processing.

Conventions. Priority uses MoSCoW; every story supporting a core KPI is Must. No story names a specific technology. Where a rule depends on an open Task 5 question, the acceptance criterion points to it instead of inventing an answer.

2. User Story Backlog (Subtask 7.1)

A. Data Collection & Automation

US-01 – Collect articles from selected sources | Must | Actor: Data Engineer | Need: N2 | KPI01

As a data engineer, I want articles from every selected Egyptian news website to be collected automatically, so that the warehouse holds a continuous record without manual copying.

- Given a source is active, when a run completes, then its new articles are stored with at least title, source, publication date and time, and original link.
- An article already stored is not stored a second time.
- If one source fails, the remaining sources are still collected in the same run.

US-02 – Run collection on a schedule | Must | Actor: System Administrator (Scheduler) | Need: N1

As a system administrator, I want collection and processing to run periodically without manual action, so that the data stays current.

- Runs start at the configured times with no manual trigger.
- Every run is recorded with start time, end time and outcome.
- A scheduled run that did not start is visible as missed, not silently absent.

US-03 – Manage the list of news sources | Should | Actor: System Administrator

As a system administrator, I want to enable or disable sources, so that the platform's coverage can change without losing history.

- A disabled source is skipped from the next run; its stored articles remain available.
- A newly enabled source is included from the next run.

US-04 – Collect responsibly | Should | Actor: Egyptian News Websites (represented by the Project Team) | Need: N6

As a news website, I want collection to follow my published terms and access restrictions, so that my content is handled appropriately.

- Each source has its access conditions reviewed and recorded before activation.
- A source whose conditions do not permit collection cannot be activated.

B. Data Quality & Attribution

US-05 – Validate and standardize articles | Must | Actor: Data Engineer | Need: N2 | Data Completeness (%)

As a data engineer, I want collected articles validated and standardized, so that the warehouse is consistent and reliable.

- Articles missing a required field are flagged and kept out of analysis, not loaded silently.
- All dates and times use one consistent format and time zone; source names are consistent.
- Data completeness (%) is reported for each run.

US-06 – Detect duplicate articles | Must | Actor: Data Engineer | Need: N2 | Duplicate Article Rate (%)

As a data engineer, I want duplicate articles detected, so that counts and timing are not inflated.

- The same article collected twice is counted once.
- Duplicate rate (%) is reported per source and period.
- Syndicated or wire-service repeats are identifiable; how they are weighted follows the decision on open question Q02.

US-07 – Preserve source attribution | Must | Actor: Egyptian News Websites / BI User | Need: N6 | Source Attribution Completeness (%)

As a news website, I want every stored article to stay attributed to its original source and link, so that content is never presented without its origin.

- Every stored article has a source and an original link.
- Every report or view that shows articles also shows their source.
- Attribution completeness (%) is available per source.

C. Source Comparison

US-08 – View articles ingested per source and period | Must | Actor: BI User | Need: N3 | KPI01

As a BI user, I want the number of ingested articles per source for a chosen period, so that I can judge how much each source contributes.

- Selecting a period shows one count per source.
- Counts match the records stored for that period.
- A source with no articles shows zero, not an error.

US-09 – Compare article volumes across sources | Must | Actor: BI User | Need: N3 | KPI01, Source Share (%) | Task 6 Story 1

As a BI user, I want to compare article counts across Egyptian news websites, so that I can identify differences in news coverage.

- Sources are shown side by side with count and share (%) of the total.
- Shares add up to 100%.
- Sources can be ranked by count.

US-10 – View source publishing frequency | Must | Actor: BI User | Need: N3 | KPI04

As a BI user, I want each source's publishing frequency and active hours, so that I can compare how often and when sources publish.

- Articles per day and per week are shown for each source.
- Active hours are shown for each source.
- Two or more sources can be compared in one view.

D. Story Coverage & Timing

US-11 – Identify stories covered by multiple sources | Must | Actor: BI User | Need: N4, N7 | KPI02 | Task 6 Story 2

As a BI user, I want to see stories covered by several sources and how many sources cover each, so that I can analyze how outlets cover similar events.

- Each story group shows its number of distinct sources.
- The list can be sorted by number of sources and filtered by a minimum number of sources.
- A story covered by one source is shown as single-source.

- The matching rule used to form story groups is documented (pending Q01).

US-12 – See which sources covered a story | Must | Actor: BI User | Need: N4 | KPI02

As a BI user, I want to open a story group and see its articles by source, so that I can compare how each source reported it.

- The view lists each article's source, title, publication time and link.
- Every article shown belongs to the selected story group.

US-13 – Compare publication times across sources | Must | Actor: BI User | Need: N5, N7 | KPI03 | Task 6 Story 3

As a BI user, I want to compare when each source published a story, so that I can examine differences in reporting timelines.

- For a story group, the first source and time are identified.
- Every other source shows its time gap from the first publication.
- Articles with missing or invalid timestamps are excluded from the gap and flagged.

E. Content Analytics

US-14 – View category distribution per source | Could | Actor: BI User | Need: N3 | KPI05

As a BI user, I want the share of each source's coverage by category, so that I can see what each source focuses on.

- Per source, category shares add up to 100%.
- Articles without a category appear as 'Uncategorized'.

F. Pipeline Monitoring

US-15 – Monitor pipeline executions | Must | Actor: System Administrator | Need: N1 | Pipeline Success Rate (%) | Task 6 Story 4

As a system administrator, I want to see the history and outcome of runs, so that I can detect problems early.

- Each run shows status, start and end time, and articles collected per source.
- Success rate (%) is shown for a chosen period.
- Failed runs are clearly distinguishable from successful ones.

US-16 – Investigate scraping failures | Must | Actor: System Administrator | Need: N1 | Scraping Failure Rate (%)

As a system administrator, I want to know which source and step failed and why, so that I can investigate and fix it.

- A failed run shows the affected source, the step, and an error description.
- Failure rate (%) is available per source and period.

G. Cross-cutting Exploration

US-17 – Filter and explore history | Must | Actor: BI User | Need: N3, N4, N5

As a BI user, I want to filter by date range, source and category and view past periods, so that I can analyze trends over time.

- Selected filters apply consistently to every KPI view.
- Historical data stays available and can be viewed as a trend over time.
- A filter with no matching data shows a clear empty result.

3. Use Case Specifications (Subtask 7.2 – draft for Ibrahim Mowafy's review)

ID	Use Case	Primary Actor	Linked Stories
UC-01	Run Scheduled Collection	Scheduler	US-01, US-02, US-04
UC-02	Validate and Standardize Articles	Data Engineer	US-05, US-06, US-07
UC-03	Group Articles into Story Groups	Data Engineer	US-11, US-12, US-13 (enabler)
UC-04	Compare News Sources	BI User	US-08, US-09, US-10, US-17
UC-05	Analyze Story Coverage	BI User	US-11, US-12, US-17
UC-06	Compare Publication Timing	BI User	US-13, US-17
UC-07	Analyze Category Distribution	BI User	US-14, US-17

UC-08	Monitor Pipeline Execution	System Administrator	US-15
UC-09	Investigate Failed Run	System Administrator	US-16
UC-10	Manage News Sources	System Administrator	US-03, US-04

Relationships: UC-01 includes UC-02; UC-03 runs after UC-02; UC-09 extends UC-08 when a failed run is selected.

UC-01 – Run Scheduled Collection

Actors: Scheduler (primary); Egyptian News Websites (supporting); System Administrator (informed). **Preconditions:** At least one source is active; a schedule is configured. **Trigger:** The scheduled time arrives.

Main flow

1. Scheduler starts a run and the platform records its start.
2. For each active source, the platform requests its latest articles.
3. The source returns articles and publication details.
4. The platform passes the collected articles to UC-02.
5. The platform records end time, articles collected per source, and outcome 'Success'.

Alternate flows

- 2a. A source has no new articles: the platform records zero for that source and continues.
- 2b. A previous run is still in progress: the new run is not started and the skip is recorded.

Failure flows

- 3a. A source is unreachable or refuses access: the platform records the failure for that source, continues with the others, and marks the run 'Partial'.
- 3b. All sources fail: the run is marked 'Failed' and appears in UC-08.

Postconditions: The run is recorded; new articles are available for processing.

UC-02 – Validate and Standardize Articles

Actors: Data Engineer (primary, rule owner); Scheduler (initiates through UC-01). **Preconditions:** Collected articles are awaiting processing.

Main flow

1. The platform checks each article for required fields.
2. It standardizes dates and times, source names and category labels.
3. It identifies duplicates and marks syndicated repeats.
4. It stores valid articles with their source and original link.
5. It records data completeness and duplicate rate for the run.

Alternate flows

- 3a. An article is a syndicated repeat: it is stored and flagged, and treated according to the Q02 decision.

Failure flows

- 1a. A required field is missing: the article is flagged as rejected with a reason and excluded from analysis.
- 2a. A date cannot be interpreted: the article is stored with the timestamp marked invalid and excluded from timing analysis.
- 4a. Storage fails: the run is marked failed at this step (see UC-09).

Postconditions: Stored articles are consistent, attributed and de-duplicated; quality figures are recorded.

UC-03 – Group Articles into Story Groups

Actors: Data Engineer (primary). **Preconditions:** Standardized articles are stored; a story-matching rule is defined (Q01).

Main flow

1. The platform selects newly stored articles.
2. It compares each against existing story groups using the matching rule.
3. A matching article joins that group; otherwise it starts a new group.
4. The platform updates each group's source count and first publication time.

Alternate flows

- 3a. An article matches more than one group: it is assigned to the best match per the rule.

Failure flows

- 2a. The rule is unavailable or fails: articles stay ungrouped and the issue is logged; coverage and timing views omit them.

Postconditions: Every processed article belongs to exactly one story group (possibly single-source).

UC-04 – Compare News Sources

Actors: BI User. **Preconditions:** Articles are stored.

Main flow

1. The BI User opens the source comparison view.
2. The user selects a period and sources (all by default).
3. The platform shows article counts, share (%) and publishing frequency with active hours per source.
4. The user ranks, filters or switches between daily and weekly views.

Alternate flows

- 2a. The user applies category or date-range filters (US-17); all figures update.

Failure flows

- 3a. No data in the selected period: the platform shows an empty-result message.
- 3b. Data cannot be retrieved: the platform shows an error and keeps the previous selection.

Postconditions: None (read-only).

UC-05 – Analyze Story Coverage

Actors: BI User. **Preconditions:** Story groups exist (UC-03).

Main flow

1. The BI User opens the story coverage view and selects a period.
2. The platform lists story groups with their number of distinct sources.
3. The user sorts or sets a minimum number of sources.
4. The user opens a group; the platform lists its articles by source with title, time and link.

Alternate flows

- 3a. The user searches or filters by source or category.

Failure flows

- 2a. No story groups in the period: the platform shows an empty-result message.
- 4a. A linked article is no longer available at the source: the stored record is shown and the link is marked unavailable.

Postconditions: None (read-only).

UC-06 – Compare Publication Timing

Actors: BI User. **Preconditions:** Story groups exist with valid timestamps.

Main flow

1. The BI User selects a story group from UC-05, or opens the timing view and selects a story.
2. The platform identifies the first-publishing source and time.
3. It shows every other source's time gap from the first publication, in chronological order.

Alternate flows

- 1a. The user compares timing across many stories for a pair of sources.

Failure flows

- 2a. Some articles have invalid timestamps: they are excluded and a notice names them.
- 2b. A story has only one source: the platform states that no comparison is possible.

Postconditions: None (read-only).

UC-07 – Analyze Category Distribution (optional)

Actors: BI User. **Preconditions:** Articles carry category labels.

Main flow

1. The BI User opens the category view and selects period and sources.
2. The platform shows each category's share (%) per source.

Alternate flows

- 2a. The user compares two or more sources side by side.

Failure flows

- 2b. Many articles are uncategorized: they appear as 'Uncategorized' and the share is stated.

Postconditions: None (read-only).

UC-08 – Monitor Pipeline Execution

Actors: System Administrator. **Preconditions:** At least one run is recorded.

Main flow

1. The administrator opens the pipeline monitoring view.
2. The platform lists runs with status, times and articles collected per source.
3. It shows success rate (%) for the selected period.
4. The administrator filters by status or period.

Alternate flows

- 2a. A run is missed: it is shown as missed with its expected time.
- 4a. The administrator selects a failed or partial run and continues to UC-09.

Failure flows

- 2b. No runs recorded: the platform states that no runs exist.

Postconditions: None (read-only).

UC-09 – Investigate Failed Run

Actors: System Administrator. **Preconditions:** A failed or partial run is selected.

Main flow

1. The platform shows the affected source(s), the failed step and an error description.
2. It shows the failure rate (%) for the affected source over the period.
3. The administrator reviews the details and records the cause outside the platform or with a note, if supported.

Alternate flows

- 3a. If re-running is supported, the administrator triggers a new run for the affected source and the outcome is recorded in UC-08.

Failure flows

- 1a. Error details were not captured: the platform states this and shows the available run information.

Postconditions: The cause is understood and any re-run is recorded.

UC-10 – Manage News Sources

Actors: System Administrator. **Preconditions:** The administrator has access to source settings.

Main flow

1. The administrator opens the source list.
2. The administrator enables or disables a source.
3. The platform confirms the change and applies it from the next run.

Alternate flows

- 2a. The administrator adds a new source with its recorded access conditions (US-04).

Failure flows

- 2b. The source's access conditions do not permit collection: activation is refused with a reason.
- 2c. A required source detail is missing: the change is rejected with a message.

Postconditions: The source list is updated; stored history is unchanged.

4. Coverage Check

4.1 Stakeholder needs (Task 6, section 2)

Need	Stakeholder	Covered by stories	Covered by use cases
N1 Monitor pipeline and identify scraping failures	Project Team	US-02, US-15, US-16	UC-01, UC-08, UC-09
N2 Complete, usable data	Project Team	US-01, US-05, US-06	UC-01, UC-02
N3 Compare volume across websites	BI Users	US-08, US-09, US-10, US-14, US-17	UC-04, UC-07
N4 Stories covered by multiple websites	BI Users	US-11, US-12, US-17	UC-03, UC-05
N5 Compare publication timing	BI Users	US-13, US-17	UC-06
N6 Correct attribution to sources	News Websites	US-04, US-07	UC-02, UC-10
N7 Understand coverage differences	General News Readers (indirect)	US-11, US-13 via BI Users	UC-05, UC-06

4.2 KPIs (Task 5 catalog and Task 6 supporting measures)

KPI	Story	Use case
KPI01 Total Articles Ingested (core)	US-01, US-08, US-09	UC-01, UC-04
KPI02 Cross-Source Story Coverage (core)	US-11, US-12	UC-03, UC-05
KPI03 Publication Time Delta (core)	US-13	UC-06
KPI04 Source Publishing Frequency (core)	US-10	UC-04
KPI05 Category Distribution Share (optional)	US-14	UC-07
Pipeline Success Rate (%)	US-15	UC-08
Scraping Failure Rate (%)	US-16	UC-09
Data Completeness (%)	US-05	UC-02
Duplicate Article Rate (%)	US-06	UC-02
Source Share (%)	US-09	UC-04
Source Attribution Completeness (%)	US-07	UC-02

All five catalog KPIs and all six Task 6 supporting measures map to at least one story and one use case. All five initial Task 6 user stories are carried over: Story 1 → US-09, Story 2 → US-11, Story 3 → US-13, Story 4 → US-15, Story 5 → US-05.

5. Findings for the Reviewer

1. **Supporting KPIs are missing from the Task 5 catalog.** Pipeline Success Rate, Scraping Failure Rate, Data Completeness, Duplicate Article Rate, Source Share and Source Attribution Completeness appear in Task 6 but not in the KPI catalog. Recommendation: add them as operational KPIs so definitions and calculation logic exist before functional requirements.
2. **Open questions that affect acceptance criteria.** Q01 (what counts as a similar story) drives US-11, US-12, US-13 and UC-03; Q02 (syndicated and wire content) drives US-06 and UC-02. Both stories are testable only once these are answered; until then the criteria reference the pending decision.
3. **Q03 (can source comparisons be written as acceptance criteria?)** Yes: US-08, US-09 and US-10 show source-level comparison expressed as testable criteria, so this question can be closed.
4. **Terminology.** 'Ingested' (KPI01) and 'collected' (Task 6) are used for the same measure, and 'story cluster' (Task 5) and 'story group' here are the same concept. Recommend fixing one term per concept in the functional requirements.
5. **Task 6 wording.** 'System maintainer' in Story 4 is replaced by System Administrator, and 'Project Team' is split into System Administrator and Data Engineer. Please confirm this split.
6. **Features marked 'if supported'.** Re-running a failed run (UC-09) and source management (US-03, UC-10) are not

confirmed project scope in the proposal. They are Should/optional until the supervisor confirms.

7. **Timing depends on data quality.** KPI03 relies on reliable publication timestamps, so US-05 and UC-02 flag invalid timestamps and exclude them from timing analysis.

6. Definition of Done

- ☒ Every stakeholder need (N1–N7) is covered by at least one story and one use case.
- ☒ Every core KPI is supported by at least one story and one use case.
- ☒ All stories have a stakeholder, need, benefit and acceptance criteria.
- ☒ All use cases have actors, preconditions, main, alternate and failure flows, and postconditions.
- ☒ Actors match the stakeholder register (with the proposed role split).
- ☒ No story assumes a specific technology.
- ☐ Ahmed Eladawy confirms stakeholder-need tracing; Bolla Kameel confirms KPI tracing.
- ☐ Ibrahim Mowafy confirms the use case drafts.
- ☐ Mustafa Mahmoud confirms coverage, scope alignment and readiness for Functional Requirements.